22BCON1068 Assignment - 2 (UNIT-2) 04/10/24

Ankit Kumar

Tutorial - 01

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- Section-A (2*5=10)

Q1. Write a RE over $\Sigma = \{a,b\}$ starting with a, but not having consecutive b's.

Ans Starting with a but not having consecutive b.

$$R.E = a^*(a + ab)^*$$

2. Write a RE over $\Sigma = \{a,b\}$ starting with 'ab'.

Ans

Starting with 'ab'.

$$RE = ab(a+b)^*$$

3. Write a RE over $\Sigma = \{a,b\}$ strings start & end with same symbol.

Ans Starting and end with same symbol

$$RE = (a(a+b)^*a + b(a+b)^*b)$$

4. Write a RE over $\Sigma = \{a,b\}$ where $|w| = 0 \pmod 2$.

Ans

Where $|w| = 0 \pmod 2$

$$RE = ((a+b)^2)^*$$

5. Write a RE over $\Sigma = \{a,b\}$ having even length of the string.

Ans

Even length of the string

$$RE = [(a+b)^2]^*$$

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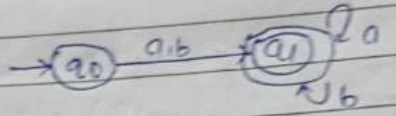
Section-B (7*3=21)

1. Prove that $aa^* + bb^*$ is the same as $(a+b)^*$.

Ans

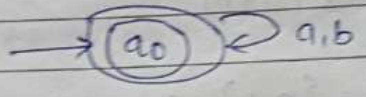
To prove $aa^* + bb^* = (a+b)^*$

DFA of L.H.S. :-

$aa^* + bb^*$ = 

here, $L_1 = \{a, b, aa, bb, aaa, bbb \ldots\}$

DFA of R.H.S :-

$(a+b)^*$ = 

$L_2 = \{\epsilon, a, b, aa, bb, \ldots\}$

Since, $L_1 \subset$ of L_2, therefore they are not equivalent.

Q2 Prove that $P + PQ^*Q = a^*bQ^*$ where $P = b + aa^*b$ and Q is any R.E.

Ans
To Prove : $P + PQ^*Q = a^*bQ^*$
Given : $P = b + aa^*b$ — ①

L.H.S:
$= P + PQ^*Q$
$= P(\epsilon + Q^*Q)$
Put $P = b + aa^*b$ from ①
$= (b + aa^*b)(\epsilon + Q^*Q)$
$= b(\epsilon + aa^*)(\epsilon + Q^*Q)$
$\because (\epsilon + aa^*) = a^*$
$\because (\epsilon + Q^*Q) = Q^*$
$= ba^*Q^*$
R.H.S proved.

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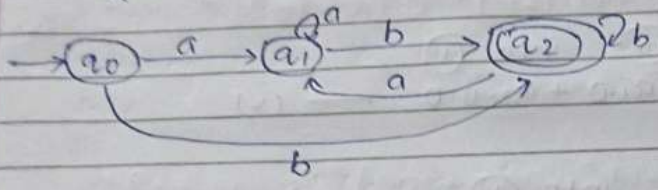
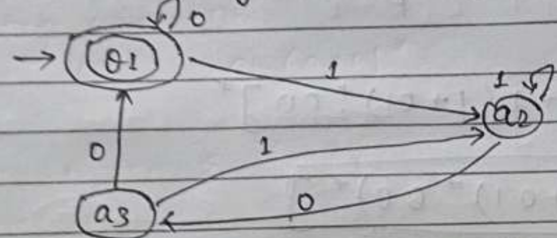
8. Construct a Finite Automata of $(ab + a^*)^* b$

Sol: $(ab+a^*)^* b$



• Section - C $(11 * 3 = 33)$

1. Write Arden's Theorem and find out regular expression for the following FA using it.



Sol: Arden's theorem states that: if P & Q are two Regular expression over an alphabet ϵ & P does not contain the empty string (ϵ) then the eqn $R = Q + RP$ has a unique soln given by $R = QP^*$.

Now,

$$q_1 = q_1.0 + \epsilon + q_3.0 \rightarrow (i)$$

$$q_2 = q_2.1 + q_1.1 + q_3.1 \rightarrow (ii)$$

$$q_3 = q_2.0 \longrightarrow (iii)$$

put eq (iii) in eq (ii)

$$q_2 = q_2.1 + q_1.1 + q_2.0.1$$

$$\underbrace{q_2}_{R} = \underbrace{q_1.1}_{Q} + \underbrace{q_2}_{R}\underbrace{(1+01)}_{P}$$

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$a_2 = (a_1 \cdot 1)(1+01)^*$ —— (iv) $R = Q + RP$
$R = QP^*$

Now,

Put (iii) into (i)

$a_1 = \epsilon + a_1 \cdot 0 + a_2 \cdot 0 \cdot 0$ —— (v)

Put (iv) into (v)

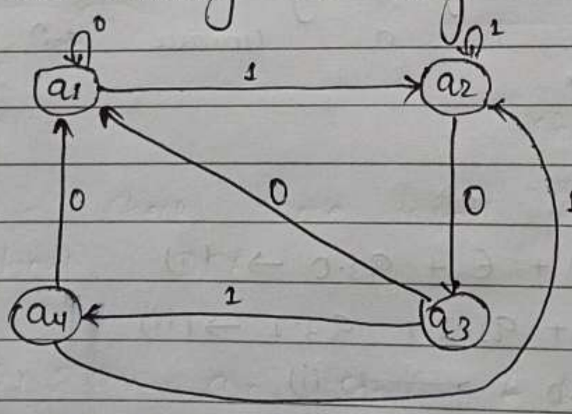
$\Rightarrow a_1 = \epsilon + a_1 \cdot 0 + (a_1 \cdot 1)(1+01)^* \cdot 00$

$\Rightarrow \underset{R}{a_1} = \underset{Q}{\epsilon} + \underset{R}{a_1}\underset{P}{(0+1(1+01)^* 0 \cdot 0)}$

$a_1 = \epsilon \cdot [0+1(1+01)^* 00]^*$

$a_1 = \boxed{0+1(1+01)^* 00)^*]}$

Q2. Write Arden's Theorem and find out regular expression for the following FA using it.



Ans

$a_1 = \epsilon + a_1 \cdot 0 + a_3 \cdot 0 + a_4 \cdot 0$ —— (i)

$a_2 = a_1 \cdot 1 + a_2 \cdot 1 + a_4 \cdot 1$ — (ii)

$a_3 = a_2 \cdot 0$ —— (iii)

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$a_4 = a_3 \cdot 1$ ——— (iv)

Put (iv) into (ii)

$a_2 = a_1 \cdot 1 + a_2 \cdot 1 + a_3 \cdot 1$ ——— (v)

Put (iii) into (v)

$a_2 = a_1 \cdot 1 + a_2 \cdot 1 + a_2 \cdot 0 \cdot 1$

$a_2 = a_1 \cdot 1 + a_2(1+01)$

R Q R P

$a_2 = a_1 \cdot 1 \, (1+01)^*$ ——— (vi)

Put (vi) into (iii)

$a_3 = (a_1 \cdot 1) \, (1+01)^*$ ——— (vii)

Put (iv) in (i)..

$a_1 = \epsilon + a_1 \cdot 0 + a_3 \cdot 0 + a_3 \cdot 1 \cdot 0$

$a_1 = \epsilon + a_1 \cdot 0 + a_3 \, (0+10)$

Put (iii) into

$a_1 = \epsilon + a_1 \cdot 0 + a_2 \cdot 0 \, (0+10)$

from (vi)

$a_1 = \varepsilon + a_1 \cdot 0 + (a_1 \cdot 1) \, (1+01)^* \, 0(0+10)$

$a_1 = \epsilon + a_1(0+1(1+01)^* \, 0(0+10))$

R Q R P

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hence ;

$$a_1 = \epsilon(0+1(1+01)^* \, 0(0+10)^*)^*$$

$$a_1 = (0+1(1+0)^* \, 0(0+10))^* \quad \text{— (viii)}$$

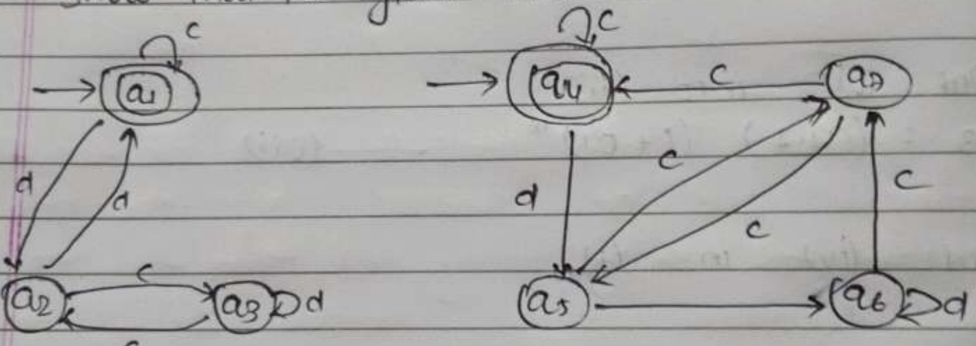
from (vii) & (viii)

$$a_3 = (0+1(1+0)^* \, 0(0+10))^* \, 1(1+01)^* \quad \text{— (ix)}$$

from (ix) & (iv)

$$a_4 = (0+1(1+0)^* \, 0(0+10)^* \, 1(1+01)^* \, 1)$$

Q3. Show that the given Automata is Equal or not.



(a) (b)

Sol^n Step 1 : Check same initial & Final state in both DFA

⇒ a_1 & a_4 are same F & I in both respectively.

Step 2 : check for conflict on transition on c & d of state (a_1, a_4).

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Conflict if : Final & intermediate
Non conflict if : Final Final & Intermediate Intermediate

$q_1 q_4$		c	d
	$q_1 q_4$	$q_1 q_4$	q_2 q_5
		F F	I I
	$q_2 q_5$	$q_3 q_1$	q_1 q_6
			F I

here Conflict pair occurs q_1 is Final & q_6 is intermediate state.

hence the given DFA are not equivalent.

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